VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (Deutz_Bardow_2021_DAC_LCA)
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Title: Life-cycle assessment of an industrial direct air capture process based on temperature–vacuum swing adsorption
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Sample: N = Commercial DAC facilities in Hinwil (waste heat) and Hellisheidi (geothermal)

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

• Process Technology: Solid-sorbent direct air capture using temperature-vacuum swing adsorption (TVSA) with amine-functionalized solid sorbents (Climeworks commercial operational data).
• Operating Conditions: Low-temperature desorption operating at 80–110 deg C (utilizing low-grade industrial waste heat or geothermal hot water/steam).
• Energy Requirements: Desorption thermal duty requires 1.5–2.5 MWh_th (5.4–9.0 GJ_th) per t-CO2 captured (baseline inventory ~11.9 GJ_th/t total including auxiliaries). Electrical duty requires 450–750 kWh_e (0.45–0.75 kWh/kg) per t-CO2 captured for fans, blowers, and vacuum pumps.
• Air Volume Throughput: ~2,000–3,000 m3 ambient air throughput per kg CO2 captured due to atmospheric dilute concentration (~420 ppm).
• Life-Cycle Removal Efficiency: Commercial plants achieve 85.4% net removal efficiency in Hinwil (Switzerland) and 93.1% in Hellisheidi (Iceland) under low-carbon geothermal energy.
• Grid Sensitivity: When powered by fossil-heavy grid electricity (~500 g CO2/kWh), net removal efficiency plummets below 40% or becomes net-positive emissions. Supply chain bottleneck: amine synthesis requires order-of-magnitude scaling for Gt-scale deployment.

VERBATIM SOURCE QUOTE / TABLE RECORD:

Current climate targets require negative carbon dioxide (CO2) emissions. Direct air capture is a promising negative emission technology, but energy and material demands lead to trade-offs with indirect emissions and other environmental impacts. Here, we show by life-cycle assessment that the commercial direct air capture plants in Hinwil and Hellisheiði operated by Climeworks can already achieve negative emissions today, with carbon capture efficiencies of 85.4% and 93.1%. The climate benefits of direct air capture, however, depend strongly on the energy source. When using low-carbon energy, as in Hellisheiði, adsorbent choice and plant construction become more important, inducing up to 45 and 15 gCO2e per kilogram CO2 captured, respectively.